

Gyo-Jin Kang

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TECHNICAL SKILLS

Languages: Python, SQL, R, Java, JavaScript, HTML/CSS, Bash

Machine Learning / AI: Machine Learning, Deep Learning, NLP, LLMs, Generative AI, MLOps

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, LangChain, FastAPI, dbt, Pandas, NumPy, GCP, React

Tools/Platforms: Snowflake, Docker, Kubernetes, AWS (EKS/ECR), GitHub Actions, Git/GitHub, ArgoCD

EXPERIENCE

Data Scientist/AI Engineer

Jan 2026 – Aug 2026

Workstream

Burnaby, BC

- Architected a multi-agent legal-compliance platform for QSR franchises using **LangChain**-orchestrated agents, incremental **dbt** models, and automated CI validation to flag statutory violations, cutting manual compliance review time by **73%**; built the analyst-facing review UI in **TypeScript/React**.
- Built a **FastAPI + LangChain** Text-to-SQL agent backed by custom **MCP** servers and **RAG**-based retrieval over **Snowflake**, cutting ad-hoc requests **30%** across **30+** CSMs; instrumented custom observability tracing into **Snowflake** for debugging and visibility.
- Built a self-evolving **LLM-as-a-Judge** evaluation loop that scores agent responses against a rubric and auto-refines its own prompts, catching regressions pre-release and cutting manual QA effort by **50%**.
- Containerized and deployed the agent suite with **Docker** and **Kubernetes** on **AWS EKS**, building **GitHub Actions** CI/CD pipelines for automated, zero-downtime weekly releases.
- Built and maintained **ELT** pipelines feeding **Snowflake** via **Hevo** ingestion across **5+** SaaS/database sources, plus a custom pipeline for **Datadog**, powering **10+** downstream dashboards.

Data Analyst

Jan 2025 – Aug 2025

British Columbia Lottery Corporation

Vancouver, BC

- Designed and ran **A/B tests** with power-analysis-driven sample sizing to optimize slot RTP configurations; winning variant lifted digital GGR by **8%**.
- Developed an **Isolation Forest** anomaly-detection service to automate fraud review, flagging **5+** fraudulent accounts monthly at a **7%** false-positive rate, cutting manual review workload.
- Built **Gaussian Mixture Model (GMM)** player segmentation across **5+** behavioral features, surfacing **4** player segments that powered targeted marketing campaigns and generated **\$1M+** incremental Q4 revenue.
- Developed a **gradient-boosted (XGBoost)** churn classifier (**AUC 0.89**) flagging at-risk players **4 weeks** in advance, paired with a **BG/NBD + Gamma-Gamma** CLV model to forecast player value — driving retention campaigns that improved player retention **40%**.

Machine Learning Engineer (Part-Time)

Sep 2024 – Jul 2025

UBC MINT

Vancouver, BC

- Built a **CNN-LSTM** hybrid model for EEG motor-imagery decoding, reaching **82%** accuracy across **4** motor movements — a **15%** improvement over a baseline SVM — via optimized spatio-temporal feature extraction.
- Engineered a **Riemannian-geometry** feature pipeline (signal covariance + tangent-space mapping, per *Barachant et al.*) and enforced strict train/test splitting to prevent leakage, correcting an inflated accuracy estimate of **90%** down to a realistic **82%** under true offline evaluation.

PROJECTS

Breast Cancer Segmentation | PyTorch

August 2024

- Built a **U-Net** model with an encoder-decoder architecture (transposed-convolution up-sampling, skip connections, batch norm, dropout) to segment malignant tumor regions in breast ultrasound images, achieving a Dice coefficient of **0.62** on **50+** test images.

EDUCATION & CERTIFICATION

University of British Columbia

Vancouver, BC

Bachelor of Science in Statistics; Economics Concentration

Sep. 2022 – May 2027 (Expected)

Deep Learning Specialization

Coursera

CNNs, RNNs, and Transformers architecture with modern optimization and regularization techniques

Mar. 2025